

International Advisory for Aerospace Travel

IAAT File *PERSEUS.debrief.24*

Perseus Rescue Mission Debriefing:

In the year 3010, a distress signal was received by the NASA Deep Space Network. Its origin was the Andromeda, a mission sent to explore the Hermes system for possible signs of life. The Hermes system had been determined to be the most likely location of life within a reachable distance of Earth. The rocky exoplanet Hermes B in particular was a good candidate for habitability.

The distress signal stated an unspecified imminent threat to the mission had been detected. It also confirmed Hermes B harbored life, the first ever discovered outside of Earth.

The crew reported multiple system failures of an unknown cause they were attempting to fix. They were making progress but requested extra resources be sent along with a replacement craft in case of total system failure. A rescue mission was assembled from several in-production spacecraft and prepared within 2 years time. The Perseus rescue mission was launched in 3013.

Given the present limitations of space travel, the rescue crew was able to make it to the Hermes system within 10 years. They made contact with what remained of Andromeda in 3023. No life was detected upon arrival. Its central module (where the research lab was housed) was found intact in orbit around Hermes B. Many of the auxiliary modules, including the living quarters, had broken off since the Andromeda crew last made contact (likely by debris impact). Most of these components of the station have yet to be located.

A search of the lab was conducted by the onboard intelligent drones accompanying Perseus. We've gathered that Andromeda's drones were likely disabled along with the other failures in the ship's systems.

Much of the resources and research of the Andromeda crew has been lost, but among the most important recoveries was a series of logs kept by the research team. These are the best insights we have into the crucial findings of Andromeda and their tragic

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disappearance. Among these logs, we also recovered personal journal entries from the crew.

The human race collectively mourns in this dark hour. These men and women traveled to a frontier of which none have been. May future generations never forget the brave crew of Andromeda and their contributions to scientific discovery.

This file contains all materials recovered by the Perseus rescue mission, organized chronologically. Let it now stand as an epitaph for those who made the ultimate sacrifice.

===== **Begin File** =====

ANDROMEDA Log 001		
Author: Captain O'Brien	Subject: Hermes Outskirts	System Status: NOMINAL
Beginning of Hermes system research logs. Today we reached the outskirts of the Hermes star system. Everything has gone as expected and the rest of the crew barely felt the time in cryosleep. The AI companions kept everything running smoothly while we were out. As expected by the observations from the team back home, we've detected 3 planets thus far, matching Hermes A, B, and C. Hermes A is a hot planet much like Venus and orbits incredibly close to its star, about 0.2 AU. Hermes C is a gas giant orbiting at about 4.1 AU. We're not close enough to get accurate readings about its composition or other details. Hermes B is what we're most interested in again as predicted prior to the mission. It's a rocky Superearth orbiting at 0.6 AU which we expect to be in the habitable zone for the system. As for the star itself, it clocks in at 0.8 Solar masses and fits within the K category as previously measured. We estimate it to be about 5 billion years old. That's all we've been able to pin down so far, but our team will be working hard for the next 6 months before we reach Hermes B.		

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We continue to think about everyone we left behind on our treasured pale blue dot.

ANDROMEDA | Log 005

Author: Captain O'Brien

Subject: Hermes B Arrival

System Status: NOMINAL

We are within sight of Hermes B and should be over there before the day is out (or whatever unit of time is standard in this region of the Milky Way). Crew morale is high and our future is looking bright.

The research team will put together a comprehensive report once the station reaches a steady orbit around the planet and we're able to set up all of our equipment properly. For now we're just enjoying the view; not quite as pretty as our home system of course.

We'll likely be up all night (now a useless figure of speech in our relativity) to make sure everything goes smoothly and to get a start as soon as possible.

ANDROMEDA | Log 006

Author: Geology Lead Osman

Subject: Hermes B Report

System Status: NOMINAL

We've now been able to get a good gauge on the planetary metrics of Hermes B. It appears to be about 3 billion years old and comes in at about 1.5 times the radius of the Earth. 40% of the planet's surface is comprised of liquid oceans and we've also detected signs of plate tectonics which has the whole crew buzzing about the possibility of life.

Something that's holding us back from that a bit is the climate of the planet. The average global temperature is around 60°C with an extremely humid atmosphere. It's certainly possible many types of life will be able to survive in this but they'd probably be very specifically adapted to it.

The planet's close orbit to Hermes seems to have started the process of tidal locking as the planet's rotation is extremely slow leading to long days and nights. Again this is causing temperature extremes.

We should have more specific readings of what's happening on the surface before tomorrow.

ANDROMEDA | Log 007

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Author: Biology Lead Chen	Subject: LIFE!	System Status: NOMINAL
We've discovered life!!! This will go down as the greatest discovery in human history. We are not alone! We've just finished setting up our equipment and our satellite imagery immediately revealed a planet teeming with an incredible diversity of life. There seem to be organisms resembling animals, plants, and fungi in an extremely similar way to Earth. It's been a day of joy and laughter and tears. The amount of work to be done here can't be understated.		

Personal Diary Engineer Williams 19/01/3005
Oh my god I can't believe it. To have come all this way and to discover this... It's too much for anyone to process. Why are we the special few who get to be here and now? Maybe because we're the only people crazy enough to travel 1 and a half parsecs away to do it. Even with all this I just can't help but miss Chloe and the kids. We all knew the implications of this journey, but maybe we were all just a little headstrong all those years ago. You'd think we'd be over it by now...

ANDROMEDA Log 021		
Author: Biology Lead Chen	Subject: Biota Overview	System Status: NOMINAL
After observing the planet for several weeks we're starting to get a good idea of the general biological makeup of the planet. For collecting data and samples we've sent our non-invasive AI drones to the surface who can work around the clock. They've been analyzing soil and water, observing organism behavior, conducting LIDAR scans, and much more. It's our goal to maintain as much of a separation from the planet as possible, at least for the foreseeable future. As much as all of us are dying to make a terrestrial landing and start walking around ourselves we have to abide by our mission protocols. The higher-ups want us to make as little contact as possible; just passive observers. (Easy for them to say, we're the ones who traveled 10 years to sit in orbit.) Despite this, we're still able to get a very good idea of the situation here. So it seems like the life on Hermes B is divided into kingdoms similar to Earth. There are mobile organisms that act similarly to animals, stationary autotrophic organisms similar to plants, and stationary heterotrophic organisms similar to fungi. There actually seems to be one species of fungi in particular that has cropped up in every research zone we've checked around the planet. This		

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includes the polar regions and areas cast in long night or day. We'll analyze a sample soon enough. I'm sure there's also a whole world of microorganisms the drones will return to the ship.

Interestingly as well some of the "animals" of this planet appear to have photosynthetic properties. Due to the long days, they're able to store up energy from the intense sunlight which helps them survive the long, cooler nights.

At the moment there doesn't seem to be any hyper-intelligent species that have reached a level of organization similar to humans. Any large structures on the planet seem to just be landmasses and natural occurrences, so far no Great Pyramids of Giza.

ANDROMEDA | Log 023

Author: BioChem Officer
Ryder

Subject: Fungi Sample
Analysis

System Status: **NOMINAL**

Our first sample of the Hermes B fungi has been returned to the ship and sequenced. It's the first organism we've been able to fully document so a new naming convention for extraterrestrial species is in order. We use Latin on Earth so we've decided for now to use Greek. Our temporary name for this fungus will be *Hermenis Pyrensus* based on the place of its origin and the Greek root for fire, for its apparent ability to spread extremely quickly.

As said before we've found this species in almost every corner of the planet and its adaptability seems to be incredible. First off, it survived the journey here. While we do try to create reasonable conditions within the sampling probes, it's not always easy. Despite the long move from the surface into orbit it seems to be doing just fine. Thriving, in fact. We've had it growing in a controlled environment and feeding it some of the Earth's plants from our botany module and it's already filled most of the terrarium we gave it.

Another thing is that it seems to do just fine through the varying temperature levels and extreme conditions on the planet. The Pyr fungus uses an extremely effective network of mycelium to suck nutrients from just about anything we give it. It seems like it might have a more dominant role in the global ecosystem than we thought. To better understand it we'll have to do more growing tests and look at the potential fossil record.

ANDROMEDA | Medbay Log 24/03/3005

Author: RN Johnson

Subject: Mild Sickness

System Status: **NOMINAL**

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Sickness Report

2 crew members have reported mild sickness.

Departments: Biology, Maintenance

Course of Action: Individuals have been quarantined for now. Inevitably we were going to make biological contact with the new planet while dealing with all of these new samples.

Hygiene and safety precautions are continuing to be observed and all scans of both the contaminant bacteria and ship apparatus have come back with low risk detected.

We will continue to monitor the situation.

ANDROMEDA | Log 024

**Author: BioChem Officer
Ryder**

Subject: Hermes B History

System Status: NOMINAL

We've made some fascinating developments in our research. We've been spending a lot of time lately using carbon dating on geologic samples we've collected and discovered some small fossil fragments here and there.

It seems as though Hermes B used to be much cooler than it is now, a lot closer to how Earth is now. Unfortunately, the lifespan of the planet seems to be rather short. The Hermes star itself is already starting to heat up and expand and Hermes B is showing signs of a runaway greenhouse. This coupled with the slowing rotation has put a tremendous strain on the planet's biosphere. Many species have died out or are not far from it. The Pyr Fungus on the other hand seems to be thriving on it. It grows incredibly fast in the hot and humid conditions and feeds on the dying organisms that aren't making it through this transitional climate period.

Life has only had a short time to develop on Hermes B, we estimate it's been around for about 2 billion years, and because of this, there hasn't been enough time for a lot of complex organisms to appear, especially ones as highly evolved as those on earth. The Pyr Fungus seems to be an exception though and has fit into a rapidly developing niche.

ANDROMEDA | Maintenance Log 20/04/3005

Author: Engineer Williams

Subject: Power Outage

System Status: WARNING

A small failure was detected in one of the power subsystems earlier. The East wing of the ship lost power for a brief bit. Shouldn't be anything to be alarmed about, these things happen fairly regularly. I swapped one of the plugs and reset the breaker and everything has

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been running smoothly since.

ANDROMEDA | Log 026

Author: Biology Lead Chen

Subject: Fungal Intelligence?

System Status: NOMINAL

This might be a huge discovery. It seems like the Pyr Fungus might actually have an extremely higher level of intelligence than we had anticipated. Their mycelium root system spans much of the continental areas of the planet. We've observed them sending signals across thousands of miles at near the speed of light.

What's even more bizarre is that large structures that initially seemed to be benign landmasses may actually be massive fungal structures they've created. They can secrete a material that hardens quickly and allows them to build large structures. They've been observed bridging across rivers and building structures to encase trees to sap all the nutrients from the dying ones. These are only a few examples but this is incredible.

It seems like they can divide their labor and specialize members of their species for different tasks. We've seen "fungal factories" they've constructed where mass amounts of reproduction and nutrient-sharing occur. Certain types of the fungi are better at reproducing, while others are best for building or fighting other species through parasitic methods or neural takeover. With this in mind it might be best for us to lay off our research until we know what we're truly dealing with.

ANDROMEDA | Medbay Log 04/05/3005

Author: RN Johnson

Subject: Illness Report

System Status: NOMINAL

Sickness Update Report

13/20 crew members are now sick with mild cold.

Again the situation is continuing to be monitored and scans have assessed a low risk.

Quarantine measures are being tightened and we ask for cooperation from all crew members at this time.

We're all very excited with the recent research conducted here but we can't let our work get in the way of the longevity and survival of the mission. Please make smart choices.

ANDROMEDA | Maintenance Log 18/05/3005

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Author: Engineer Williams	Subject: Power Outages	System Status: WARNING
The power outages have started to get worse. It's even starting to affect our communication with the drones down on the surface. It seems we actually might have a serious issue I need to take care of. I was hoping to not have to do this but I'm gonna head into the main engine compartment via the maintenance hatch to find the root of the problem.		

ANDROMEDA Captain Alert System		
Author: Captain O'Brien	Subject: URGENT	System Status: WARNING
This is an urgent report to all crew members. Officer Williams has been reported missing. He was last seen early yesterday when he left to troubleshoot our power issues. A search party will be formed shortly to find him. Just in case, stick together for the time being.		

ANDROMEDA Captain Alert System		
Author: First Officer Kahnwald	Subject: URGENT	System Status: CRITICAL
I've found Captain O'Brien unresponsive in the bridge. It's been reported all maintenance entrances are stuck shut with some type of strong resin-like material. Fungal masses have been reported throughout the ship and it's likely we may have a station-wide infestation. A distress signal has been sent to home. For now our best course of action is to not have any direct contact with the ship. We will hold.		

===== End of File =====